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Ian Zhang

Curriculum Vitae

Contact Information

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Education

PhD in Statistics, University of Toronto *Sep 2026 -*

- Advisors: Joshua Speagle, Thibault Randrianarisoa

MSc in Statistics, University of Toronto *Sep 2025 - Jun 2026*

- Advisor: Nancy Reid

Honours BSc in Statistics/Mathematics, University of Toronto *Sep 2021 - Jun 2025*

- Advisor: Jun Young Park
- Graduated with high distinction

Publications and Preprints

†: Corresponding Author; *: Equal contribution

Published

1. Michael A. Kouritzin, **Ian Zhang**†, Jyoti Bhadana, Seoyeon Park. Markov processes for enhanced deepfake generation and detection[J]. *AIMS Mathematics*, 2026, 11(4): 11731-11759. doi: `10.3934/math.2026483`

In Progress

1. Aditya Khan, **Ian Zhang** *2026*
   Parametric Hotspot Analysis with the Getis-Ord Statistic

2. Hanlong Chen*, **Ian Zhang***, . . ., Howard Chertkow, . . ., Malcolm Binns *2026*
   Reproducible Dementia Subtyping through Cross-Validated Joint Hyperparameter Optimization of PCA dimensionality and k in Unsupervised Clustering

3. **Ian Zhang**†, Thibault Randrianarisoa *2026*
   Likelihood Tempering to Mitigate Prior Dominance in Variational Posteriors for Bayesian Neural Networks

Research Experience

Graduate Research Assistant, Baycrest Institute/University of Toronto *2025-Present*
Supervisor: Malcolm Binns

Summer Research Trainee, McGill University *2025*
Supervisor: Jianguo (Jeff) Xia

Research Assistant, University of Alberta *2023-2024*
Supervisor: Mike Kouritzin

Teaching

Teaching Assistant

UofT Statistical Sciences Research Program (UTSSRP) *Summer 2026*
STA437: Methods of Multivariate Data *Winter 2026*
MAT137: Calculus with Proofs *Fall 2025 & Winter 2026*
STA302: Methods of Data Analysis I *Fall 2025*

Presentations

Conference Talks

2026 Joint Statistical Meetings *2026*
Will be presenting *RAMA: Redundancy-Aware Model Assessment for PCA-Based Clinical Subtype Discovery.*

2026 SSC Annual Meeting in Hamilton *2026*
Presented *Likelihood Tempering to Mitigate Prior Dominance in Variational Posteriors for Bayesian Neural Networks* in the New Frontiers in Bayesian Analysis session.

2025 SSC Annual Meeting in Saskatoon *2025*
Presented *Markov Processes for Enhanced Deepfake Generation and Detection* in the Probability session.

Third Joint SIAM/CAIMS Annual Meetings *2025*
Presented *Markov Processes for Enhanced Deepfake Generation and Detection* in the Machine Learning and Stochastic Processes session.

Posters

The Thirteenth Annual Canadian Statistics Student Conference *2025*
Presented *Comparing Exact and Approximate Inference for Spatial Autocorrelation Tests: A Power Analysis.*

Honours and Awards

Anna & Alex Beverly Memorial Fellowship *2025*
Awarded by University College (University of Toronto) for general proficiency during my undergraduate degree.

NSERC Undergraduate Student Research Award (x2) *2023, 2024*
Awarded by the University of Alberta to research Markov process methods for deepfake generation and detection.

Jack Zwaigenbaum Scholarship *2022*
Awarded by University College (University of Toronto) for academic excellence in my first year of undergraduate studies.

Dean's List Scholar (x4) *2022, 2023, 2024, 2025*
Awarded by the Faculty of Arts and Science (University of Toronto) for maintaining academic excellence every year of my undergraduate degree.

Grants

CSSC Travel Grant *2025*
Awarded to full-time Canadian students presenting a poster or giving a talk at the Canadian Statistics Student Conference.

Services

Conferences

Session chair, 2025 Third Joint SIAM/CAIMS Annual Meetings
Chaired the "Machine Learning and Stochastic Processes" session.

Other

Citizenship: Canadian
Languages Spoken: English (native), Mandarin (native), Japanese (intermediate)
Memberships: Statistical Society of Canada